

# Paper 1 - Set 10 - Solutions

Generated practice paper: answer key and worked solutions.

## Answer key

| Question | Answer | Topic                                      |
|----------|--------|--------------------------------------------|
| 1        | D      | Binomial expansion                         |
| 2        | B      | Exponentials and logarithms                |
| 3        | F      | Angles, polygons, congruence and vectors   |
| 4        | F      | Sequences and series                       |
| 5        | A      | Indices and surds                          |
| 6        | D      | Integration                                |
| 7        | F      | Polynomials, factor and remainder theorems |
| 8        | F      | Pythagoras, mensuration and 3D             |
| 9        | B      | Sequences and series                       |
| 10       | A      | Binomial expansion                         |
| 11       | B      | Trigonometric functions and equations      |
| 12       | B      | Number, units and estimation               |
| 13       | A      | Graphs of functions                        |
| 14       | C      | GCSE algebra                               |
| 15       | A      | Exponentials and logarithms                |
| 16       | E      | Functions and transformations              |
| 17       | B      | Polynomials, factor and remainder theorems |
| 18       | C      | Simultaneous equations and inequalities    |
| 19       | C      | Differentiation                            |
| 20       | F      | Indices and surds                          |

## Worked solutions

### Question 1 BIN-07 · Binomial expansion · spec MM2.4

Expand  $(1 - 2x)^4$  and hence find the coefficient of  $x^2$  in  $(1 + 3x)(1 - 2x)^4$ .

**Answer: D**

$(1 - 2x)^4 = 1 - 8x + 24x^2 - 32x^3 + 16x^4$ . Coefficient of  $x^2$  in the product:  $1 \cdot 24 + 3 \cdot (-8) = 0$ .

#### Full method - how and why

A product of a binomial expansion with a linear factor. Only two products can give  $x^2$ , so collect just those.

1 Expand  $(1 - 2x)^4$ : coefficients  $\binom{4}{r}(-2)^r$  give  $1 - 8x + 24x^2 - 32x^3 + 16x^4$ .

why:  $\binom{4}{1}(-2) = -8$  and  $\binom{4}{2}(-2)^2 = 6 \times 4 = 24$ ; remember to raise the  $-2$  to the power as well.

2 To make  $x^2$  in the product, pair  $1 \times 24x^2$  and  $3x \times (-8x)$ .

why: Degrees must add to 2: (0, 2) and (1, 1) are the only options.

3 Add:  $24 + (-24) = 0$ .

why: The two contributions cancel exactly - the answer is 0, not 'no term'.

#### Where it goes wrong

- Taking the  $x^2$  coefficient of  $(1 - 2x)^4$  as +24 but forgetting the  $3x \times (-8x)$  contribution, giving 24.
- Sign error making  $(-2)^2 = -4$ .

**Takeaway** For a coefficient in a product, list the degree pairs that add to the target and sum their contributions.

### Question 2 LOG-18 · Exponentials and logarithms · spec MM5.2

If  $\log_a b = 2$  and  $\log_b c = 3$ , find  $\log_a c$ .

**Answer: B**

$b = a^2$  and  $c = b^3 = a^6$ , so  $\log_a c = 6$ .

#### Full method - how and why

A chain of logarithms. Convert each statement to exponential form and substitute.

1  $\log_a b = 2$  means  $b = a^2$ .

why:  $\log_a b = c \Leftrightarrow a^c = b$ .

2  $\log_b c = 3$  means  $c = b^3$ .

why: Same conversion with the next pair.

3 Substitute:  $c = (a^2)^3 = a^6$ .

why: Powers of powers multiply.

4 Therefore  $\log_a c = 6$ .

why: Read the exponent straight off. (In general  $\log_a c = \log_a b \times \log_b c = 2 \times 3$ .)

#### Where it goes wrong

- Adding the values to get 5.
- Reading  $(a^2)^3$  as  $a^5$ .

**Takeaway** Logarithms chain by multiplication:  $\log_a c = \log_a b \times \log_b c$ .

### Question 3 GEO-01 · Angles, polygons, congruence and vectors · spec M5.2

The interior angle of a regular polygon is  $156^\circ$ . How many sides does it have?

**Answer: F**

Exterior angle  $24^\circ$ ;  $360/24 = 15$  sides.

#### Full method - how and why

Work with the **exterior** angle: they always sum to  $360^\circ$ , which makes the arithmetic trivial.

- 1 Exterior angle  $= 180^\circ - 156^\circ = 24^\circ$ .

why: Interior and exterior angles at a vertex lie on a straight line.

- 2 The exterior angles of any polygon sum to  $360^\circ$ , and for a regular polygon they are all equal.

why: Walking once around the shape turns you through a full circle.

- 3 Number of sides  $= \frac{360}{24} = 15$ .

why: Divide the total turn by the turn per vertex.

#### Where it goes wrong

- Using  $\frac{360}{156}$  or  $\frac{156}{24}$ .
- Using the interior-angle-sum formula and then having to solve  $\frac{180(n-2)}{n} = 156$  - correct but slower.

**Takeaway** For regular polygons, go via the exterior angle:  $n = \frac{360}{\text{exterior}}$ .

### Question 4 SEQ-10 · Sequences and series · spec MM2.3

A geometric series has first term 2 and common ratio  $r > 0$ . The sum of the first three terms is 14. Find the sum of the first five terms.

**Answer: F**

$$2 + 2r + 2r^2 = 14 \Rightarrow r^2 + r - 6 = 0 \Rightarrow r = 2. S_5 = 2(2^5 - 1) = 62.$$

#### Full method - how and why

The first three terms give a quadratic in  $r$ ; the positivity condition picks one root.

- 1 Write the sum of three terms:  $2 + 2r + 2r^2 = 14$ .

why: Terms are  $a$ ,  $ar$ ,  $ar^2$  with  $a = 2$ .

- 2 Divide by 2 and rearrange:  $r^2 + r - 6 = 0$ .

why: Dividing first keeps the numbers small.

- 3 Factorise:  $(r + 3)(r - 2) = 0$ , so  $r = 2$  or  $r = -3$ . The condition  $r > 0$  gives  $r = 2$ .

why: The question's restriction exists precisely to discard the negative root.

- 4 Sum five terms:  $S_5 = 2 \frac{2^5 - 1}{2 - 1} = 2 \times 31 = 62$ .

why:  $2^5 = 32$ , so the bracket is 31.

#### Where it goes wrong

- Using  $r = -3$  and getting  $S_5 = 2 \times \frac{-243 - 1}{-4} = 122$ .
- Adding two more terms by hand and slipping:  $2 + 4 + 8 + 16 + 32 = 62$  is a good check.

**Takeaway** Conditions like ' $r > 0$ ' are there to select between roots - apply them before continuing.

**Question 5** IND-14 · Indices and surds · spec MM1.2

The number  $\frac{\sqrt{50} + \sqrt{18}}{\sqrt{8}}$  is equal to

**Answer: A**

$$\sqrt{50} = 5\sqrt{2}, \sqrt{18} = 3\sqrt{2}, \sqrt{8} = 2\sqrt{2}. \text{ So } \frac{8\sqrt{2}}{2\sqrt{2}} = 4.$$

**Full method - how and why**

A quotient of surds that collapses completely. Simplify each root first and the  $\sqrt{2}$ 's cancel.

1 Simplify:  $\sqrt{50} = 5\sqrt{2}, \sqrt{18} = 3\sqrt{2}, \sqrt{8} = 2\sqrt{2}$ .

why: Every number is (square)  $\times 2$ , so every root is a multiple of  $\sqrt{2}$  - which is why the fraction will be rational.

2 Add the numerator:  $5\sqrt{2} + 3\sqrt{2} = 8\sqrt{2}$ .

why: Like terms again -  $\sqrt{2}$  behaves as a common factor.

3 Divide:  $\frac{8\sqrt{2}}{2\sqrt{2}} = 4$ .

why: The  $\sqrt{2}$  cancels top and bottom, leaving plain arithmetic.

**Where it goes wrong**

- Working out  $\sqrt{50} + \sqrt{18} = \sqrt{68}$ , which is wrong and leads nowhere.
- Cancelling only the numbers and leaving a stray  $\sqrt{2}$ , giving  $4\sqrt{2}$ .

**Takeaway** Simplify every surd to the form  $k\sqrt{m}$  before adding or dividing; common surd factors then cancel visibly.

**Question 6** INT-20 · Integration · spec MM7.2

Evaluate  $\int_{-1}^2 |x| dx$ .

**Answer: D**

Two triangles:  $\frac{1}{2} \cdot 1 \cdot 1 + \frac{1}{2} \cdot 2 \cdot 2 = \frac{1}{2} + 2 = \frac{5}{2}$ .

**Full method - how and why**

A modulus integrand. Split at the corner, or - faster - read the areas off as triangles.

1 Note  $|x|$  changes formula at  $x = 0$ , which is inside  $[-1, 2]$ .

why: Split at every point where the modulus switches branch.

2 On  $[-1, 0]$ :  $|x| = -x$ , and the graph is a triangle with base 1 and height 1, area  $\frac{1}{2}$ .

why: Sketching  $y = |x|$  makes the integral a pair of triangles - no antiderivative needed.

3 On  $[0, 2]$ :  $|x| = x$ , a triangle with base 2 and height 2, area 2.

why:  $\frac{1}{2} \times 2 \times 2 = 2$ .

4 Add:  $\frac{1}{2} + 2 = \frac{5}{2}$ .

why: Both pieces are above the axis, so both count positively.

**Where it goes wrong**

- Integrating  $x$  straight through and getting  $\frac{4}{2} - \frac{1}{2} = \frac{3}{2}$  - an offered option.
- Treating the left piece as negative.

**Takeaway**  $|x|$  is never negative: split at the corner, and use triangle areas when the graph is piecewise linear.

**Question 7** POL-02 · Polynomials, factor and remainder theorems · spec MM1.6

$(x + 2)$  is a factor of  $2x^3 + ax^2 - 5x - 6$ . Find the value of  $a$ .

**Answer: F**

Factor Theorem:  $f(-2) = -16 + 4a + 10 - 6 = 4a - 12 = 0$ , so  $a = 3$ .

**Full method - how and why**

'Is a factor' means remainder zero, so the Factor Theorem turns the statement into a single equation for  $a$ .

- 1  $(x + 2)$  vanishes at  $x = -2$ .

why: Write  $x + 2 = x - (-2)$  if the sign is confusing; the root is the value that makes the bracket zero.

- 2 Substitute:  $f(-2) = 2(-8) + a(4) - 5(-2) - 6 = -16 + 4a + 10 - 6$ .

why: Careful with the odd power:  $(-2)^3 = -8$ , while  $(-2)^2 = +4$ .

- 3 Simplify:  $4a - 12$ .

why:  $-16 + 10 - 6 = -12$ .

- 4 Set to zero (factor  $\Rightarrow$  remainder zero):  $4a = 12$ , so  $a = 3$ .

why: That is precisely the Factor Theorem.

**Where it goes wrong**

- Using  $x = 2$  instead of  $x = -2$ , which gives  $a = -3$  - an offered option.
- Sign slips on  $(-2)^3$  and  $-5(-2)$ ; write each term out separately.

**Takeaway** Factor  $(x + k)$  means substitute  $x = -k$  and set the result to zero.

**Question 8** MEN-14 · Pythagoras, mensuration and 3D · spec M5.16

A cylindrical tin of radius 4 cm and height 10 cm is full of water. The water is poured into an empty cylindrical jug of radius 5 cm. How deep is the water in the jug?

**Answer: F**

$$\pi \cdot 16 \cdot 10 = \pi \cdot 25 \cdot h \Rightarrow h = \frac{160}{25} = 6.4.$$

**Full method - how and why**

Conservation of volume: the water keeps its volume, so equate the two cylinder volumes.

- 1 Volume in the tin:  $\pi r^2 h = \pi(16)(10) = 160\pi$ .

why:  $r = 4$  so  $r^2 = 16$ .

- 2 In the jug the same volume occupies  $\pi(25)h$ .

why: Radius 5, unknown depth  $h$ ; the  $\pi$  will cancel.

- 3 Equate and solve:  $160\pi = 25\pi h \Rightarrow h = \frac{160}{25} = 6.4$  cm.

why: The wider jug means a shallower depth - a good sanity check.

**Where it goes wrong**

- Using the radii rather than their squares, giving 8 cm - an offered option.
- Expecting the depth to increase because the jug is 'bigger'.

**Takeaway** Pouring conserves volume: equate  $\pi r_1^2 h_1 = \pi r_2^2 h_2$ ; a wider container gives a shallower depth.

### Question 9 SEQ-09 · Sequences and series · spec MM2.2

How many terms of the arithmetic series  $3 + 7 + 11 + \dots$  are needed for the sum to first exceed 500?

**Answer: B**

$$S_n = \frac{n}{2}(6 + 4(n - 1)) = n(2n + 1). \quad n = 15: 465; \quad n = 16: 528 > 500. \text{ So } 16.$$

#### Full method - how and why

An inequality on a sum. Form  $S_n$ , then find the first integer  $n$  that works - by trial, since  $n$  must be a whole number.

- 1 Identify  $a = 3$ ,  $d = 4$  and form the sum:  $S_n = \frac{n}{2}[6 + 4(n - 1)] = \frac{n}{2}(4n + 2) = n(2n + 1)$ .

why: Simplifying to  $n(2n + 1)$  makes the trial values much easier to compute.

- 2 Test  $n = 15$ :  $15 \times 31 = 465$ , which is less than 500.

why: Estimate first:  $2n^2 \approx 500$  gives  $n \approx 15.8$ , so start near there.

- 3 Test  $n = 16$ :  $16 \times 33 = 528 > 500$  ✓.

why: The sum is increasing, so the first  $n$  that exceeds 500 is the answer.

- 4 Answer: 16 terms.

why: Check the one below to be sure it fails - that is what 'first exceeds' demands.

#### Where it goes wrong

- Solving  $2n^2 + n = 500$  and rounding down to 15.
- Forgetting that  $n$  must be a positive integer and quoting a decimal.

**Takeaway** For 'first exceeds' questions, solve approximately then test the two neighbouring integers.

### Question 10 BIN-05 · Binomial expansion · spec MM2.4

Evaluate  $\binom{10}{3} - \binom{10}{7} + \binom{9}{2}$ .

**Answer: A**

$$\binom{10}{3} = \binom{10}{7}, \text{ so the first two cancel; } \binom{9}{2} = 36.$$

#### Full method - how and why

A short calculation made trivial by the symmetry of binomial coefficients.

- 1 Use the symmetry  $\binom{n}{r} = \binom{n}{n-r}$ : since  $3 + 7 = 10$ ,  $\binom{10}{3} = \binom{10}{7}$ .

why: Choosing 3 objects from 10 is the same as choosing which 7 to leave behind.

- 2 So the first two terms cancel, leaving  $\binom{9}{2}$ .

why: Spotting the cancellation avoids computing 120 twice.

- 3 Evaluate  $\binom{9}{2} = \frac{9 \times 8}{2} = 36$ .

why:  $\binom{n}{2} = \frac{n(n-1)}{2}$ .

#### Where it goes wrong

- Computing all three coefficients and mis-adding, e.g.  $120 - 120 + 36$  done as 156.
- Thinking  $\binom{10}{3} - \binom{10}{7}$  is  $\binom{10}{-4}$  or similar.

**Takeaway**  $\binom{n}{r} = \binom{n}{n-r}$ : look for pairs whose lower entries add to  $n$ .

**Question 11** TRG-16 · Trigonometric functions and equations · spec MM4.6

Given that  $\sin x + \cos x = 0$  and  $0 \leq x < 2\pi$ , find the sum of all possible values of  $x$ .

**Answer: B**

$$\tan x = -1: x = \frac{3\pi}{4} \text{ or } \frac{7\pi}{4}. \text{ Sum} = \frac{10\pi}{4} = \frac{5\pi}{2}.$$

**Full method - how and why**

Convert a sum of sin and cos into a single tan equation by dividing.

- 1 Rearrange:  $\sin x = -\cos x$ .  
why: Move one term across.
- 2 Divide by  $\cos x$ :  $\tan x = -1$ . (Check  $\cos x = 0$  separately: then  $\sin x = 0$  too, impossible.)  
why: Dividing by  $\cos x$  is safe only after ruling out  $\cos x = 0$  - and here that case gives a contradiction.
- 3 Solve on  $[0, 2\pi)$ :  $\tan x = -1$  at  $x = \frac{3\pi}{4}$  and  $x = \frac{3\pi}{4} + \pi = \frac{7\pi}{4}$ .  
why: Reference angle  $\frac{\pi}{4}$ , negative tangent in quadrants 2 and 4.
- 4 Sum:  $\frac{3\pi}{4} + \frac{7\pi}{4} = \frac{10\pi}{4} = \frac{5\pi}{2}$ .  
why: Add over the common denominator and simplify.

**Where it goes wrong**

- Finding only one solution and answering  $\frac{3\pi}{4}$  - an offered option.
- Forgetting to justify dividing by  $\cos x$  (here it is fine, but the habit matters).

**Takeaway**  $a \sin x + b \cos x = 0$  becomes  $\tan x = -\frac{b}{a}$ ; check the  $\cos x = 0$  case before dividing.

**Question 12** NUM-16 · Number, units and estimation · spec M1.1

Water flows through a pipe at 1.2 litres per second. How long does it take to fill a tank of volume  $3.6 \text{ m}^3$ ? ( $1 \text{ m}^3 = 1000$  litres.)

**Answer: B**

3600 litres at 1.2 litres/s takes  $3000 \text{ s} = 50$  minutes.

**Full method - how and why**

A rate problem with a unit conversion. Convert the volume to litres, divide by the rate, then convert the time.

- 1 Convert the volume:  $3.6 \text{ m}^3 = 3.6 \times 1000 = 3600$  litres.  
why: The conversion is given in the question.
- 2 Divide by the rate:  $\frac{3600}{1.2} = 3000$  seconds.  
why: Volume divided by volume-per-second gives seconds.
- 3 Convert to minutes:  $\frac{3000}{60} = 50$  minutes.  
why: The options are in minutes, so finish the conversion.

**Where it goes wrong**

- Leaving the answer as 3000 seconds and picking 500 seconds by a slip.
- Dividing by 1.2 as if it were 12, giving 5 minutes.

**Takeaway** Make the units match before dividing, and convert the answer into the units the options use.

**Question 13** GRA-02 · Graphs of functions · spec MM8.7

The graphs of  $y = x^2 - 2x$  and  $y = k$  meet at exactly two points. Find the complete set of values of  $k$ .

**Answer: A**

$y = (x - 1)^2 - 1$  has minimum  $-1$ ; a horizontal line above the minimum meets the parabola twice.

**Full method - how and why**

A horizontal line meeting a parabola twice: the line must be above the minimum.

- 1 Complete the square:  $x^2 - 2x = (x - 1)^2 - 1$ , so the minimum value is  $-1$  at  $x = 1$ .

*why:* The turning point governs how many times a horizontal line can cut the curve.

- 2 A line  $y = k$  meets the parabola twice when  $k$  is **strictly above** the minimum.

*why:* At  $k = -1$  the line is tangent at the vertex: exactly one point. Below it: none.

- 3 So  $k > -1$ .

*why:* Strict inequality, because equality gives only one intersection.

**Where it goes wrong**

- Including  $k = -1$  and choosing  $k \geq -1$  - offered as a distractor.
- Using the  $y$ -intercept  $0$  instead of the minimum value.

**Takeaway** Count intersections with a horizontal line by sliding it against the turning point; equality gives tangency.

**Question 14** ALG-14 · GCSE algebra · spec M4.16

The sum of two numbers is 12 and the difference of their squares is 48. Find the larger number.

**Answer: C**

$$x^2 - y^2 = (x - y)(x + y) = 12(x - y) = 48 \Rightarrow x - y = 4. \text{ So } x = 8, y = 4.$$

**Full method - how and why**

A sum and a difference of squares. Factorising the difference of squares links them without solving for both numbers.

- 1 Factorise:  $x^2 - y^2 = (x - y)(x + y)$ .

*why:* This is the identity that connects the two given facts.

- 2 Substitute the known sum:  $48 = (x - y)(12)$ , so  $x - y = 4$ .

*why:* The sum  $x + y = 12$  is given directly.

- 3 Solve the pair  $x + y = 12$ ,  $x - y = 4$ : adding gives  $2x = 16$ , so  $x = 8$  (and  $y = 4$ ).

*why:* Adding the two equations eliminates  $y$ .

- 4 The larger number is 8.

*why:* Check:  $64 - 16 = 48$  ✓.

**Where it goes wrong**

- Answering 4 (the smaller number or the difference).
- Trying to solve  $x^2 - y^2 = 48$  by trial.

**Takeaway**  $x^2 - y^2 = (x - y)(x + y)$  converts a difference of squares into a linear pair.



**Question 15** LOG-01 · Exponentials and logarithms · spec MM5.2

Find the value of  $\log_2 40 - \log_2 5 + \log_2 \frac{1}{4}$ .

**Answer: A**

$$\log_2 \left( \frac{40}{5} \cdot \frac{1}{4} \right) = \log_2 2 = 1.$$

**Full method - how and why**

Combine everything into a single logarithm using the laws, then evaluate.

1 Combine the first two:  $\log_2 40 - \log_2 5 = \log_2 \frac{40}{5} = \log_2 8$ .

why: A difference of logs (same base) is the log of the quotient.

2 Add the third:  $\log_2 8 + \log_2 \frac{1}{4} = \log_2 \left( 8 \times \frac{1}{4} \right) = \log_2 2$ .

why: A sum of logs is the log of the product. Note  $\log_2 \frac{1}{4}$  is negative, so this step reduces the total.

3 Evaluate:  $\log_2 2 = 1$ .

why:  $\log_a a = 1$  for any valid base.

**Where it goes wrong**

- Treating  $\log_2 \frac{1}{4}$  as  $+2$  rather than  $-2$  and answering 5.
- Dividing the logarithms instead of subtracting them.

**Takeaway** log turns  $\times$  into  $+$  and  $\div$  into  $-$ ; combine into one log before evaluating.

**Question 16** FUN-15 · Functions and transformations · spec MM8.2

Which of the following is the equation of the graph obtained by stretching  $y = x^3 - 3x$  parallel to the  $x$ -axis with scale factor  $\frac{1}{2}$ ?

**Answer: E**

A stretch parallel to the  $x$ -axis with factor  $\frac{1}{2}$  gives  $y = f(2x) = (2x)^3 - 3(2x) = 8x^3 - 6x$ .

**Full method - how and why**

A horizontal stretch of factor  $\frac{1}{2}$  replaces  $x$  by  $2x$  - the reciprocal of the factor.

1 A stretch parallel to the  $x$ -axis with scale factor  $k$  gives  $y = f\left(\frac{x}{k}\right)$ ; with  $k = \frac{1}{2}$  that is  $y = f(2x)$ .

why: Inside changes act inversely: to squash the graph to half its width you must double the input.

2 Substitute:  $y = (2x)^3 - 3(2x)$ .

why: Replace every  $x$  in the formula, including the one in the linear term.

3 Simplify:  $8x^3 - 6x$ .

why:  $(2x)^3 = 8x^3$  - the coefficient is cubed too.

**Where it goes wrong**

- Replacing only the first  $x$ , giving  $8x^3 - 3x$  - an offered option.
- Using  $y = f(x/2)$  and getting  $\frac{x^3}{8} - \frac{3x}{2}$ , which is a stretch of factor 2.

**Takeaway** Stretch factor  $k$  parallel to the  $x$ -axis  $\Rightarrow$  replace  $x$  by  $x/k$ ; substitute into **every** occurrence.

**Question 17** POL-12 · Polynomials, factor and remainder theorems · spec MM1.6

When  $x^3 - 3x^2 + 2x + 7$  is divided by  $x^2 + 1$ , what is the remainder?

**Answer: B**

$x^3 - 3x^2 + 2x + 7 = (x^2 + 1)(x - 3) + (x + 10)$ : expanding gives  $x^3 - 3x^2 + x - 3 + x + 10$ . Remainder  $x + 10$ .

**Full method - how and why**

Division by a quadratic again: remainder is linear. Comparing coefficients is fastest because the divisor has no  $x$  term.

- 1 Write  $x^3 - 3x^2 + 2x + 7 = (x^2 + 1)(x + c) + (ax + b)$ .  
why: Cubic divided by quadratic gives a linear quotient and a linear remainder.
- 2 Expand the right:  $x^3 + cx^2 + x + c + ax + b$ .  
why:  $(x^2 + 1)(x + c) = x^3 + cx^2 + x + c$ .
- 3 Compare  $x^2$ :  $c = -3$ .  
why: Only the quotient contributes to  $x^2$ .
- 4 Compare  $x$ :  $1 + a = 2$ , so  $a = 1$ . Compare constants:  $c + b = 7$ , so  $b = 7 + 3 = 10$ .  
why: Work down the powers; each comparison introduces exactly one new unknown.
- 5 Remainder:  $x + 10$ .  
why: Degree 1 < degree 2 ✓.

**Where it goes wrong**

- Forgetting the  $+x$  that  $(x^2 + 1)(x)$  contributes, giving  $a = 2$  and the remainder  $2x + 10$ .
- Using the Remainder Theorem with  $x^2 + 1$  - it has no real roots, so that route needs complex numbers and is off-specification.

**Takeaway** Comparing coefficients handles any divisor; only use root substitution when the divisor has real roots.

**Question 18** SIM-16 · Simultaneous equations and inequalities · spec MM1.5

Find the complete set of values of  $x$  for which  $\frac{x-2}{x+1} \leq 0$ .

**Answer: C**

A fraction is  $\leq 0$  when numerator and denominator have opposite signs (or the numerator is 0):  $x - 2 \leq 0$  and  $x + 1 > 0$  gives  $-1 < x \leq 2$ ;  $x - 2 \geq 0$  and  $x + 1 < 0$  is impossible.  $x = -1$  is excluded as the fraction is undefined.

**Full method - how and why**

A rational inequality. Never multiply by  $x + 1$  (its sign is unknown) - analyse the sign of the fraction instead.

- 1 Find the critical values: numerator zero at  $x = 2$ , denominator zero at  $x = -1$ .  
why: The fraction can only change sign where the numerator or denominator vanishes.
- 2 Test the three regions. For  $x < -1$  (say  $x = -2$ ):  $\frac{-4}{-1} = 4 > 0$ . For  $-1 < x < 2$  (say  $x = 0$ ):  $\frac{-2}{1} = -2 < 0$ . For  $x > 2$  (say  $x = 3$ ):  $\frac{1}{4} > 0$ .  
why: One test value per region settles the sign throughout that region.
- 3 Include  $x = 2$  (the fraction is 0, and  $\leq 0$  allows it) but exclude  $x = -1$  (undefined).  
why: The two endpoints behave differently: a zero numerator is allowed by  $\leq$ ; a zero denominator never is.
- 4 Answer:  $-1 < x \leq 2$ .  
why: Note the mixed brackets - strict at  $-1$ , inclusive at 2.

**Where it goes wrong**

- Multiplying by  $x + 1$  to get  $x - 2 \leq 0$ , i.e.  $x \leq 2$ , which wrongly includes everything below  $-1$ .
- Writing  $-1 \leq x \leq 2$  and including a value where the expression is undefined.

**Takeaway** For  $\frac{A}{B} \leq 0$ , use a sign chart on  $A$  and  $B$ ; a zero denominator is always excluded, a zero numerator is included when the inequality is not strict.

**Question 19** DIF-13 · Differentiation · spec MM6.1

$f(x)$  is a polynomial with  $f'(x) = (x - 2)^2(x + 1)$ . Which of the following is true?

**Answer: C**

$f'$  changes sign from negative to positive at  $x = -1$  (local minimum). At  $x = 2$  the factor  $(x - 2)^2$  does not change sign, so  $f'$  stays positive: a stationary point but not a turning point (a point of inflexion).

**Full method - how and why**

Reading a factorised derivative. A squared factor does **not** change the sign, so it gives a stationary point that is not a turning point.

- The stationary points are the roots of  $f'$ :  $x = -1$  and  $x = 2$ .  
why:  $f'(x) = 0$  at both.
- Near  $x = -1$ :  $(x - 2)^2 > 0$  always, so the sign of  $f'$  is the sign of  $(x + 1)$  - negative then positive.  
why: A sign change from  $-$  to  $+$  means the function falls then rises: a local minimum.
- Near  $x = 2$ :  $(x + 1) > 0$  there, and  $(x - 2)^2 \geq 0$ , so  $f' \geq 0$  on both sides - no sign change.  
why: A repeated (even-power) factor touches zero without crossing, so the gradient does not change sign.
- Conclusion: local minimum at  $x = -1$ ; stationary but **not** a turning point at  $x = 2$ .  
why: That is a stationary point of inflexion.

**Where it goes wrong**

- Assuming every root of  $f'$  is a turning point.
- Using  $f''(2) = 0$  to conclude 'maximum' or 'minimum' - a zero second derivative is inconclusive.

**Takeaway** Odd-power factors of  $f'$  change sign (turning points); even-power factors do not (inflexions).

**Question 20** IND-18 · Indices and surds · spec MM1.2

The expression  $\frac{3}{7 - 2\sqrt{3}} - \frac{3}{7 + 2\sqrt{3}}$  simplifies to

**Answer: F**

Common denominator  $(7 - 2\sqrt{3})(7 + 2\sqrt{3}) = 49 - 12 = 37$ . Numerator:  $3(7 + 2\sqrt{3}) - 3(7 - 2\sqrt{3}) = 12\sqrt{3}$ .

**Full method - how and why**

A difference of two rationalisable fractions. Using a single common denominator is faster than rationalising each one.

- Use the common denominator  $(7 - 2\sqrt{3})(7 + 2\sqrt{3}) = 49 - 12 = 37$ .  
why: The two denominators are conjugates, so their product is automatically rational - no separate rationalising needed.
- Numerator:  $3(7 + 2\sqrt{3}) - 3(7 - 2\sqrt{3}) = 3 \times 4\sqrt{3} = 12\sqrt{3}$ .  
why: When you subtract conjugates the rational parts cancel and the surd parts double - so only the surd survives.
- Result:  $\frac{12\sqrt{3}}{37}$ .  
why: Already rationalised, since the denominator is an integer.

**Where it goes wrong**

- $(2\sqrt{3})^2 = 4 \times 3 = 12$ , not 6; using 6 gives denominator 43.
- Adding the numerators instead of subtracting, giving  $\frac{42}{37}$  - which is offered.

**Takeaway** A difference of conjugate fractions leaves only the surd part; their product denominator is rational from the start.